

SAKTHIVEL A

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SUMMARY

Senior Full Stack Engineer with 4+ years of experience building and scaling enterprise-grade, high-performance web applications across Java/Spring Boot and MERN stacks. Proven ownership delivering greenfield products to production, including AI-driven recommendation platforms and data-driven admin systems. Strong expertise in Micro-Frontend architecture (Module Federation), microservices, AWS, and performance optimization, with a consistent track record of reliable delivery and collaboration across global teams.

TECHNICAL SKILLS

- **Languages:** Java (8/11/17), JavaScript (ES6+), TypeScript, Python, SQL
- **Frontend:** React.js, Next.js, Redux, Ant Design, Material UI, HTML5, CSS3, Tailwind CSS
- **Architecture:** Micro-Frontend Architecture, Module Federation, Microservices, Event-Driven Architecture, Reusable Component Design
- **Backend & APIs:** Node.js, Express.js, Fastify, Spring Boot, Spring MVC, RESTful APIs, GraphQL, Kafka
- **Databases:** MongoDB, MySQL, Redis, DynamoDB
- **ORM & Data Access:** Hibernate, JPA, Spring Data JPA, Sequelize, Prisma, Knex, JDBC
- **Security & Auth:** Keycloak, FusionAuth, JWT, OAuth, Spring Security, RBAC
- **Cloud:** AWS (EC2, S3, RDS, Lambda, IAM, CloudWatch, SNS, SQS, Route53), GCP
- **DevOps:** Git, GitHub, GitHub Actions, Jenkins, Docker, Kubernetes, OpenShift, CI/CD Pipelines
- **Testing:** JUnit, Mockito, Mocha
- **Practices:** Agile/Scrum, Code Reviews, Performance Optimization
- **AI Tools:** Cursor, GitHub Copilot

WORK EXPERIENCE

Senior Software Engineer, Photon Interactive Private Limited

Sep 2025 – Present

- Led an 8-member cross-functional team (frontend, backend, QA, BA, and Scrum Master) in the end-to-end development of Tenetic, a geography-based advertisement planning platform, owning frontend and backend architecture.
- Took full ownership of the frontend, restructuring the codebase and driving open production bugs down from 450 to 0.
- Implemented AWS integration (S3) for secure storage and data handling within the platform.
- Built data pipelines to ingest client inputs (products, target regions, holidays, demographics) and generate ad-timing and placement recommendations across U.S. states.
- Designed scalable APIs in Node.js/Fastify with MySQL for structured campaign data and MongoDB for flexible client inputs, integrating Python-based recommendation logic following an MVC architecture.
- Collaborated daily with QA, BA, Scrum Master, and the engineering manager on sprint delivery, partnering with the LLM/Python AI team during production support.

Associate Software Engineer, Kaay Labs Private Limited

May 2022 – Aug 2025

- Led development and enhancement of large-scale enterprise admin platforms (Pie Factory, Mouser Electronics) across pricing, data processing, and workflow management domains.
- Led a 5-member team on Pie Factory to build a configuration-driven UI and deliver an early POC, including dynamic chart visualization and print/export functionality.
- Architected Micro-Frontend systems on Mouser Electronics using Module Federation, collaborating with a 15-member cross-functional team for independent deployment and seamless integration of multiple frontend applications.
- Optimized API and data-loading performance to reliably handle 100K+ row Excel/XL datasets in production.
- Owned critical UI modernization initiatives, migrating legacy UI frameworks (Ant Design v3 → v8), improving performance, UI consistency, and long-term maintainability.

- Provided round-the-clock production support as the primary escalation point for high-severity issues, while owning code reviews and mentoring junior engineers across distributed teams.

PROJECT EXPERIENCE

Tenetic *(Geography-Based Advertisement Planning Platform)*

Tech Stack: Next.js, React.js, TypeScript, Node.js, Fastify, Python, MySQL, MongoDB, AWS, REST, Redux, Tailwind CSS

- Built a geography-based ad planning platform recommending optimal ad timing and placement by U.S. state, demographic, holiday, and product.
- Used Python for AI-driven recommendation logic, integrated with a Node.js/Fastify backend following an MVC architecture.
- Designed MySQL schemas for structured campaign/product data, with MongoDB for flexible, less-structured client inputs.
- Built the frontend in Next.js/React with Redux and Tailwind CSS for dashboard visualization of recommendations.
- Deployed on AWS, ensuring reliable delivery under tight client timelines.

Pie Factory *(Admin & Client Portal with Analytics Dashboard & Report Management)*

Tech Stack: React.js, Next.js, TypeScript, Node.js, Express.js, MySQL, AWS, REST, GraphQL, JWT, Prisma

- Built an admin and client portal with real-time analytics dashboards and report generation, using React.js/Next.js with GraphQL for efficient data fetching.
- Designed REST/GraphQL APIs in Node.js/Express.js, using Prisma as the ORM layer for MySQL.
- Implemented JWT-based authentication with role-based access control for admin and client user types.
- Deployed on AWS, optimizing dashboard performance for large historical datasets.

Mouser Electronics *(Admin Portals & Pricing Tool)*

Tech Stack: Java, Spring Boot, React.js, TypeScript, Node.js, MySQL, AWS, Kafka, Hibernate, JPA, Microservices, Keycloak, Module Federation, OpenShift, Kubernetes, Jenkins, Docker

- Designed enterprise admin portals on a Java Spring Boot microservices backend with Hibernate/JPA, paired with a Module Federation Micro-Frontend in React.js.
- Built real-time market-based price comparison tools, integrating Spring Boot services with Node.js frontend APIs.
- Implemented Kafka-based event streaming between microservices for asynchronous order and pricing updates, improving decoupling and fault tolerance.
- Secured the platform with Keycloak across microservices for authentication and role-based access control.
- Containerized and deployed services via Docker/Kubernetes on OpenShift, with Jenkins CI/CD pipelines.
- Resolved complex cross-service and cross-module dependency issues, ensuring stability across federated modules in production.

DataBP *(Multi-Tenant Admin Panel)*

Tech Stack: React.js, TypeScript, Node.js, MongoDB, MySQL, AWS, Knex, Keycloak

- Built a scalable multi-tenant admin system in React/Node.js to manage client orders, track progress, and control order lifecycles.
- Used Knex as the query builder for MySQL, with MongoDB for flexible client-specific configuration data.
- Secured multi-tenant access via Keycloak, ensuring data isolation and role-based access control across clients.
- Optimized application performance and scalability to support concurrent enterprise users and diverse client environments.

EDUCATION

Bachelor of Engineering (CSE)

Jun 2015 – Apr 2019

Anna University College of Engineering

CERTIFICATES

- Full Stack Developer Internship – Namaste JavaScript